

Assignment Submission Instructions

Chat UI Development
CampusPe

Assignment Date: April 04, 2026 | Maximum Marks: 110
Technologies: HTML, CSS, JavaScript, jQuery & Bootstrap 5
Submission: Zip file — YourName_ChatUI.zip

Assignment Overview

In this assignment, you will create a modern, responsive chat user interface similar to Claude or ChatGPT. You will build this using only HTML, CSS, JavaScript, jQuery, and Bootstrap 5. No backend required — focus on creating a beautiful, interactive, and user-friendly chat experience.

Learning Objectives

- Create responsive layouts using CSS Flexbox and Grid
- Style modern UI components with CSS custom properties
- Implement smooth animations and transitions
- Handle user interactions with JavaScript and jQuery
- Build reusable UI components
- Create mobile-responsive designs
- Apply UX best practices for chat interfaces

Prerequisites

- Basic knowledge of HTML5 and semantic markup
- Understanding of CSS3 including selectors, box model, positioning
- Basic JavaScript knowledge including variables, functions, DOM manipulation
- Familiarity with jQuery syntax
- Text editor such as VS Code, Sublime Text, or similar
- Modern web browser with developer tools

What You Will Build

You will create a complete chat interface with the following components:

- Welcome screen with suggestion cards
- Message display area with user and AI message bubbles
- Animated typing indicator
- Input area with auto-resizing textarea
- Send button with state management
- Sidebar with chat history
- Mobile-responsive navigation
- Smooth scrolling and animations

Assignment Tasks

This assignment is divided into 4 progressive tasks. Complete them in order as each builds upon the previous one.

Task 1: HTML Structure and Basic Layout (25 points)

Part 1.1: Create HTML Structure (10 points)

- Create a proper HTML5 document with semantic tags

- Include links to Bootstrap 5, Font Awesome, and Google Fonts
- Create a container for the entire app
- Build the messages container with proper wrapper
- Create the input area with textarea and buttons
- Add typing indicator structure
- Use semantic HTML such as header, main, section

Part 1.2: Welcome Screen (8 points)

- Create a centered welcome section
- Add a large icon or logo
- Create 4 suggestion cards in a grid layout
- Each card should have an icon, title, and description
- Make cards clickable for future functionality

Part 1.3: Input Area Structure (7 points)

- Create input container with textarea
- Add attach button with paperclip icon
- Add send button with paper plane icon
- Include footer text below input
- Structure should support auto-resize functionality

Task 2: CSS Styling and Design (35 points)

Part 2.1: CSS Variables and Global Styles (8 points)

- Define CSS custom properties for colors
- Set up variables for spacing and sizing
- Create variables for shadows and transitions
- Apply global reset for margin, padding, box-sizing
- Set base font family and colors

Part 2.2: Message Bubbles Styling (12 points)

- Style user messages with right-aligned and custom background
- Style AI messages with left-aligned and different background
- Create circular avatars with gradient backgrounds
- Add proper spacing and padding to message content
- Style message header with avatar, name, time
- Add border-radius for rounded corners
- Different colors for user vs AI avatars

Part 2.3: Input Area Styling (8 points)

- Style textarea with proper padding and border
- Add focus state with accent color border
- Style buttons with hover effects
- Create smooth transitions for all interactive elements
- Add box-shadow to input container
- Style send button with disabled state

Part 2.4: Animations and Effects (7 points)

- Create keyframes for fade-in animation on new messages
- Add typing indicator animation with bouncing dots
- Implement hover effects on cards and buttons
- Add smooth transitions to all interactive elements
- Create slide-in animation for welcome screen

Task 3: JavaScript and jQuery Functionality (25 points)**Part 3.1: Message Display (10 points)**

- Create addMessage function that accepts text and sender
- Function should create message HTML and append to chat
- Add timestamp to each message with current time
- Different styling for user vs assistant messages
- Implement scrollToBottom function
- Auto-scroll when new message is added

Part 3.2: Input Handling (8 points)

- Enable or disable send button based on input
- Handle Enter key to send message with Shift+Enter for new line
- Clear input after sending message
- Auto-resize textarea as user types
- Prevent sending empty messages

Part 3.3: Mock AI Responses (7 points)

- Create array of sample AI responses
- Implement sendMessage function
- Show typing indicator when processing
- Simulate delay of 1 to 2 seconds before AI response
- Hide typing indicator and show response
- Hide welcome screen after first message

Task 4: Sidebar and Advanced Features (15 points)**Part 4.1: Create Sidebar (8 points)**

- Create sidebar with fixed width of 260px
- Add New Chat button at top
- Create scrollable chat history section
- Add user profile section at bottom
- Style with dark background color
- Add hover effects on history items

Part 4.2: Mobile Responsive Design (7 points)

- Add hamburger menu button hidden on desktop
- Sidebar slides in from left on mobile
- Create overlay for sidebar on mobile

- Adjust message bubble width on mobile
- Stack suggestion cards vertically on mobile
- Test on different screen sizes from 320px to 1920px

Bonus Tasks (Extra 10 points)

Complete any of these optional features for extra credit:

- Dark mode toggle with smooth theme transition — 4 points
- Message formatting support for code blocks, bold, italic — 3 points
- Typing animation that types out AI response letter by letter — 4 points
- Export chat as text file using JavaScript Blob API — 3 points
- Custom scrollbar styling — 2 points
- Add sound effects for message send and receive — 2 points

Grading Rubric

Component	Points	Criteria
Task 1: HTML	25	Semantic HTML, proper structure, all required elements
Task 2: CSS	35	Beautiful design, variables, animations, responsiveness
Task 3: JavaScript	25	All functions working, smooth interactions, no errors
Task 4: Sidebar & Mobile	15	Sidebar working, responsive on all devices
Code Quality	10	Clean code, proper indentation, meaningful names
Design Aesthetics	10	Professional look, color harmony, good UX
Bonus Features	10	Additional features — optional
TOTAL	110	100 points required for full credit

Design Requirements

Your chat UI must have:

- Clean, modern appearance inspired by Claude or ChatGPT style
- Smooth animations on all interactions
- Clear visual distinction between user and AI messages
- Responsive design that works on mobile, tablet, desktop
- Accessible color contrast meeting WCAG AA standards
- Professional typography using Google Fonts
- Intuitive user experience

Submission Guidelines

What to Submit

- Complete HTML file with all components

- CSS file with all styling — well-organized and commented
- JavaScript file with all functionality
- Screenshots showing your UI on different devices
- README file with brief description and how to run
- Zip file named: YourName_ChatUI.zip

File Structure

Your submission should follow this structure:

```

YourName_ChatUI/
  index.html – main HTML file
  css/style.css – your custom styles
  js/chat.js – your JavaScript code
  screenshots/ – desktop.png, mobile.png, tablet.png
  README.md – setup instructions
  
```

Testing Checklist

Before submission, verify each item below:

- Messages appear correctly when sent
- User and AI messages are visually different
- Send button is disabled when input is empty
- Send button is enabled when there is text
- Enter key sends message; Shift+Enter creates new line
- Typing indicator shows and hides correctly
- Auto-scroll works when new messages appear
- Textarea auto-resizes as user types
- Suggestion cards are clickable
- Welcome screen hides after first message
- Sidebar appears and hides on mobile
- Layout is responsive on all screen sizes
- No console errors in browser developer tools

Detailed Evaluation Criteria

Excellent (90–100%)	Good (75–89%)	Needs Work (<75%)
All features work perfectly	Most features work	Many features missing
Beautiful, professional design	Good design, minor issues	Poor visual design
Smooth animations throughout	Some animations present	No or broken animations
Perfect mobile responsiveness	Works on most devices	Not responsive
Clean, well-organized code	Readable code	Messy, hard to read
No errors in console	Minor console warnings	Multiple errors

Tips for Success

- Start with HTML structure first, then CSS, then JavaScript
- Use browser developer tools to debug and test
- Test your code frequently — do not wait until the end
- Keep your code organized with proper indentation
- Use meaningful variable and function names
- Comment your code to explain complex parts
- Look at the provided reference UI for inspiration
- Test on different browsers such as Chrome, Firefox, Safari
- Use CSS variables for easy color theme changes
- Mobile-first approach: design for mobile, then scale up
- Keep animations subtle — do not overdo it
- Ask questions early if you are stuck

Common Mistakes to Avoid

- Not testing on mobile devices
- Forgetting to clear input after sending message
- Not handling empty messages
- Using inline styles instead of CSS file
- Not commenting code
- Copy-pasting code without understanding it
- Inconsistent spacing and indentation
- Not using CSS variables for colors
- Forgetting to include jQuery library
- Not testing in multiple browsers

Helpful Resources

- Bootstrap 5 Documentation: <https://getbootstrap.com/docs/5.3/>
- jQuery API Documentation: <https://api.jquery.com/>
- Font Awesome Icons: <https://fontawesome.com/icons>
- Google Fonts: <https://fonts.google.com/>
- CSS-Tricks tutorials: <https://css-tricks.com/>
- MDN Web Docs: <https://developer.mozilla.org/>
- Can I Use browser compatibility: <https://caniuse.com/>
- ColorHunt color palettes: <https://colorhunt.co/>

Recommended Timeline

Day	Tasks	Time Estimate
Day 1	Complete Task 1 — HTML Structure	2–3 hours

Day	Tasks	Time Estimate
Day 2–3	Complete Task 2 — CSS Styling	4–5 hours
Day 4	Complete Task 3 — JavaScript	3–4 hours
Day 5	Complete Task 4 — Sidebar and Mobile	2–3 hours
Day 6	Testing, Screenshots, Documentation	1–2 hours
Day 7	Bonus features (optional) and Final review	2–3 hours
Total	Estimated time	12–18 hours

Frequently Asked Questions

Q: Can I use a different CSS framework?

A: Yes, but Bootstrap 5 is recommended for easier responsiveness.

Q: Do I need to make the AI actually work?

A: No! Just use mock responses. Focus is on the UI only.

Q: Can I work with a partner?

A: No, this is an individual assignment.

Q: What if I cannot finish all tasks?

A: Submit what you have. Partial credit will be given.

Q: Can I use additional libraries?

A: Yes, as long as you can explain what they do.

Q: How will code quality be graded?

A: Based on organization, comments, naming, and no errors.

Q: Is the design pixel-perfect required?

A: No, but it should look professional and polished.

Q: Can I change colors?

A: Yes! Feel free to customize colors professionally.

Questions and Support

- Post in the course discussion forum
- Attend office hours: Monday and Wednesday 4–5 PM
- Email: instructor@tattvacodelabs.com
- Join Discord for peer support
- Check the FAQ section above first

Academic Integrity

You must write your own code. While you can reference documentation and tutorials, copying code directly from classmates or online sources is plagiarism. You should be able to explain every line of code in your submission.

Good Luck! Have fun building a beautiful chat interface.